# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 6 — MATHS · NCERT-TYPE UNIT TEST SET 2

Chapters 1–5: **Patterns in Mathematics, Lines and Angles, Number Play, Data Handling and Presentation, Prime Time** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

## SECTION — A (Objective Type Questions) 15 × 1 = 15

*Choose the correct option and write it in your answer book.*

**1.** What is the sum of the first 15 natural numbers $1 + 2 + 3 + \cdots + 15$? 1

**(a)** 121 **(b)** 135 **(c)** 120 **(d)** 105

**2.** If you face North and turn clockwise through 90 degrees, what further clockwise turn (in degrees) brings you back to facing North? 1

**(a)** 360 **(b)** 270 **(c)** 280 **(d)** 260

**3.** The number of steps to reach a palindrome by repeatedly reversing and adding, starting from 57: 1

**(a)** 4 **(b)** 1 **(c)** 3 **(d)** 2

**4.** The total number of students in the table is: 1

| Sport | Students |
|---|---|
| Hockey | 25 |
| Tennis | 30 |
| Cricket | 15 |
| Football | 25 |
| Kabaddi | 50 |

**(a)** 150 **(b)** 145 **(c)** 140 **(d)** 155

**5.** The number of prime numbers between 10 and 30 is: 1

**(a)** 5 **(b)** 8 **(c)** 7 **(d)** 6

**6.** The numbers 1, 4, 9, 16, 25, ... are called square numbers. The 16th one is: 1

**(a)** 257 **(b)** 256 **(c)** 272 **(d)** 240

**7.** 4 rays start from a point O. How many angles (counting pairs of rays) do they make with each other? 1

**(a)** 10 **(b)** 5 **(c)** 7 **(d)** 6

**8.** The digit sum of 4567 is: 1

**(a)** 23 **(b)** 22 **(c)** 21 **(d)** 31

**9.** Find the range (largest - smallest) of the data: 42, 39, 31, 45, 44, 34, 40, 11 1

**(a)** 35 **(b)** 36 **(c)** 34 **(d)** 33

**10.** The prime factorisation of 72 has how many prime factors (counting repeats)? 1

**(a)** 6 **(b)** 7 **(c)** 4 **(d)** 5

**11.** In the pattern $1, 3, 6, 10, \ldots$ what is the 9th term? 1

**(a)** 45 **(b)** 44 **(c)** 54 **(d)** 46

**12.** A protractor shows the arm of an angle at $35^\circ$ on the inner scale. The corresponding reading on the outer scale is: 1

**(a)** 135 **(b)** 235 **(c)** 155 **(d)** 145

**13.** A supercell: the number in the middle cell is the sum of its two neighbours. If neighbours are 13 and 18, the number is: 1

**(a)** 31 **(b)** 32 **(c)** 30 **(d)** 41

**14.** In the data 1, 5, 2, 2, 4, 4, 4, 4, 4, 3, 2, 2, 5, 3, 4, 2, the number which occurs most often is: 1

**(a)** 3 **(b)** 4 **(c)** 5 **(d)** 6

**15.** Using the Sieve of Eratosthenes, which is the first prime whose multiples we cross out after crossing multiples of 2? 1

**(a)** 4 **(b)** 1 **(c)** 3 **(d)** 5

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 20–30 words each.*

**16.** How many lines are needed to join all pairs among 8 points (a complete graph)? 2

**17.** What is the angle between the hands of a clock at 3:00? 2

**18.** Do one Kaprekar step on 2961. 2

**19.** A bar graph is drawn with a scale of 1 cm = 5 students for the data below. Find the height of each bar. 2

| Item | Number |
|---|---|
| Football | 15 |
| Hockey | 25 |
| Kabaddi | 25 |

**20.** Write two prime numbers whose sum is 24. 2

## SECTION — C (Short Answer Type) — 5 × 3 = 15

*Answer in about 40–60 words each.*

**21.** Find the number of steps taken by the Collatz sequence starting at 13 to reach 1 and the largest number in it. 3

**22.** A scale 1 cm = 10 units is used. Bars are of height 4 cm, 7 cm and 5 cm. Find the values, and the difference between the largest and the smallest. 3

**23.** Draw a factor tree (describe) for 96 and write its prime factorisation. 3

**24.** Look at the sequence $5, 7, 9, 11, 13$. Find the pattern and the 10th term. 3

OR

How many right angles make 180 degrees? What is the angle after 3 such turns? 3

**25.** A number is made by writing 7 5 times. Find its digit sum and say whether it is divisible by 3. 3

OR

Which scale is better to draw a bar graph for values 20, 40, 60: 1 cm = 1 unit or 1 cm = 10 units? Give the height of the largest bar with your chosen scale. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**26.** Find all prime numbers between 1 and 40 using the Sieve of Eratosthenes idea. State which primes you cross out multiples of. **5**

**OR**

In the Virahanka sequence $1, 2, 3, 5, 8, \ldots$ the 5th and 6th terms are 8 and 13. Find the next three terms and add the first 5 terms. **5**

**27.** Find the angle between the hands at 2:20 using: hour hand at $60^\circ$ plus $10^\circ$ and minute hand at $120^\circ$. Name the type of the (smaller) angle. **5**

**OR**

Show with working that adding 121 and 232 (both palindromes) gives 353. Is the sum a palindrome? Make a palindrome using the reverse-and-add process starting from 29. **5**

**END OF QUESTION PAPER** — **Total: 50 Marks**

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